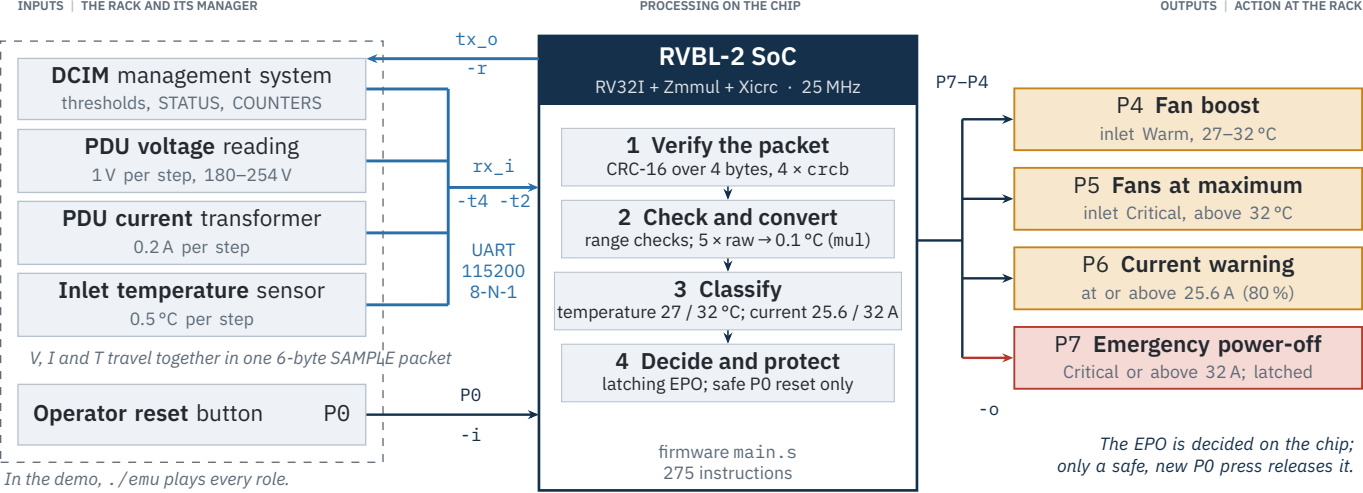
INPUTS | THE RACK AND ITS MANAGER
PROCESSING ON THE CHIP
OUTPUTS | ACTION AT THE RACK
DCIM management system
thresholds, STATUS, COUNTERS
PDU voltage reading
1 V per step, 180–254 V
PDU current transformer
0.2 A per step
Inlet temperature sensor
0.5 °C per step
V, I and T travel together in one 6-byte SAMPLE packet
Operator reset button
P0
In the demo, ./emu plays every role.
tx_o
-r
rx_i
-t4 -t2
UART
115200
8-N-1
P0
-i
RVBL-2 SoC
RV32I + Zmmul + Xicrc · 25 MHz
1 Verify the packet
CRC-16 over 4 bytes, 4 × crcb
2 Check and convert
range checks; 5 × raw → 0.1 °C (mul)
3 Classify
temperature 27 / 32 °C; current 25.6 / 32 A
4 Decide and protect
latching EPO; safe P0 reset only
firmware main.s
275 instructions
P7–P4
-o
P4 Fan boost
inlet Warm, 27–32 °C
P5 Fans at maximum
inlet Critical, above 32 °C
P6 Current warning
at or above 25.6 A (80 %)
P7 Emergency power-off
Critical or above 32 A; latched
The EPO is decided on the chip;
only a safe, new P0 press releases it.